

CPTAC3 expression 代謝物資料取最大值、中位數

檔案：CPTAC3_PDC000XXX_GBM_metabolite_expression_matrix_byCaseID.xlsx

搭配使用：代謝物 ID+名稱、Pubchem、HMDB

工作表介紹：

1. modified_result_max：整理最大值的工作表(使用 excel 合併彙整得出的工作表)
2. tmp1：代謝物名稱排序後待處理的資料(保留，當在 tmp2 誤刪時，從 tmp1 查詢、使用)
3. tmp2：代謝物名稱排序後待處理的資料(整理完的代謝物資料可刪減)
4. modified_result_median_tmp：套用公式將同一個代謝物取中位數的暫時工作表
5. modified_result_median：整理中位數的工作表

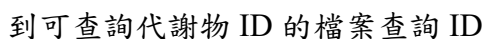
方法步驟摘要：

在整理最大值、中位數的工作表時，將代謝物名稱加上代謝物 ID，由於整理完第一次後得到的工作表並非最終結果，會再經由排序代謝物 ID，將相同 ID 的排在一起，進行二次整理，整理完後才能進行後續分析

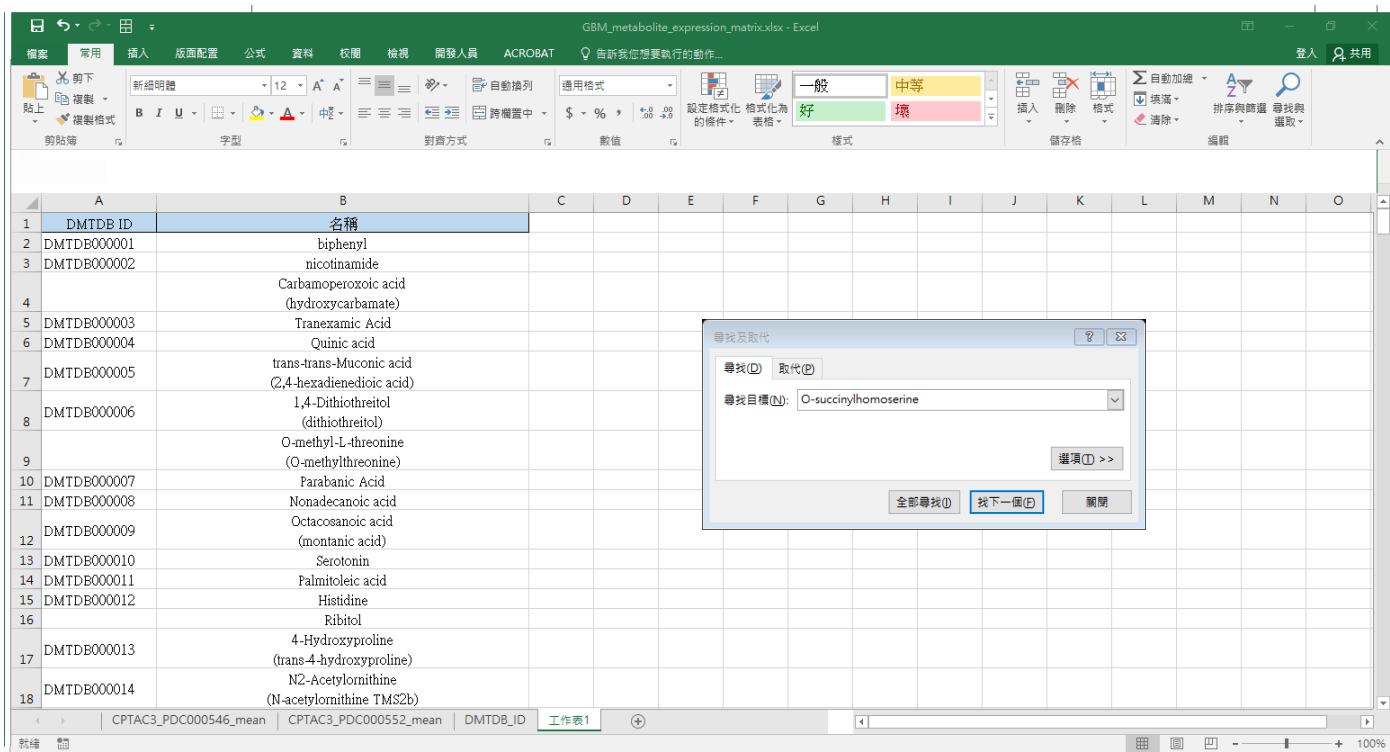
使用到的 excel 技巧：

1. 合併彙整：“資料”->“合併彙算”(取最大值)
2. 公式：=MEDIAN(範圍)

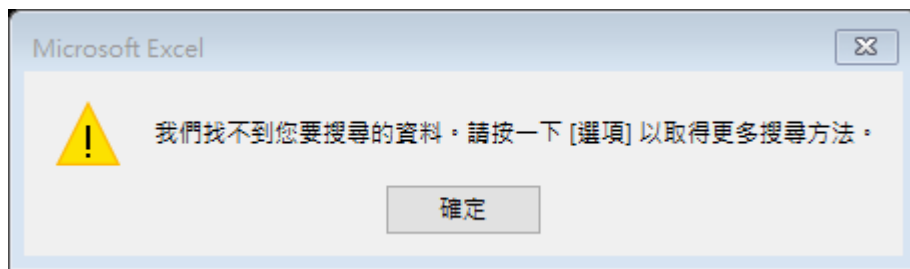
到 tmp2 中選取代謝物名稱



1. ctrl + F 中的查詢多按幾次，有些代謝物名稱很像，須找到最符合的代謝物名稱
2. 若是查不到，到 Pubchem 的網站上查詢



搜尋不到：



到 Pubchem 上搜尋(<https://pubchem.ncbi.nlm.nih.gov/>)

PubChem Search has been updated
You may continue to [use the previous version](#) for a limited time. For questions or feedback, visit our [contact page](#). Recent changes are described in the [release notes](#).

O-succinylhomoserine

Treating this as a text search.

BEST COMPOUND MATCH

 O-succinyl-L-homoserine; 1492-23-5; O-Succinylhomoserine; O4-Succinyl-L-homoserine; (2S)-2-amino-4-(3-carboxypropanoyloxy)butanoic acid; ...
Compound CID: 439406
MF: C₈H₁₃NO₆ MW: 219.19 g/mol
IUPAC Name: (2S)-2-amino-4-(3-carboxypropanoyloxy)butanoic acid
SMILES: C(COC(=O)CCC(=O)O)[C@@H](C(=O)O)N
InChIKey: GNISQJGXJDKDJ-YFKPBYRVSA-N
InChI: InChI=1S/C8H13NO6/c9-5(8(13)14)3-4-15-7(12)2-1-6(10)11/h5H,1-4,9H2,(H,10,11)(H,13,14)/t5-m/0/s1
Create Date: 2004-09-16
[Summary](#) [Similar Structures Search](#) [Related Records](#) [PubMed \(MeSH Keyword\)](#)

ALL DATA COLLECTIONS TEXT SEARCH

Compounds	Substances	RefChem	Genes	Proteins	Literature	Patents
(1)	(18)	(1)	(9)	(43)	(26)	(50)

Searching chemical names and synonyms including IUPAC names and InChIKeys across the compound collection. Note that annotations text from compound summary pages is not searched.

<https://pubchem.ncbi.nlm.nih.gov/> Download

若是搜尋 Pubchem 的結果有很多筆(在最上面的搜尋結果不是 BEST COMPOUND MATCH)或是搜尋不到，就將 tmp2 中的該代謝物的列刪除，進行下一個代謝物名稱的搜尋與整理

點進去 Pubchem 的化合物頁面後，選擇右方的 Names and identifiers

pubchem.ncbi.nlm.nih.gov/compound/439406

An official website of the United States government [Here is how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

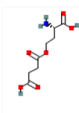
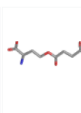
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COMPOUND

O-Succinylhomoserine

PubChem CID	439406
Structure	 
Molecular Formula	C ₈ H ₁₃ NO ₆
Synonyms	O-succinyl-L-homoserine 1492-23-5 O-Succinylhomoserine O4-Succinyl-L-homoserine (2S)-2-amino-4-(3-carboxypropanoyloxy)butanoic acid View More...

Cite Download

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- 3 Names and Identifiers**
- 4 Chemical and Physical Properties
- 5 Spectral Information
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- 9 Pharmacology and Biochemistry
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- 13 Taxonomy
- 14 Classification

1. 在這一區往下滑找到 HMDB 的網頁連結

pubchem.ncbi.nlm.nih.gov/compound/439406#section=Names-and-Identifiers

PubChem O-Succinylhomoserine (Compound)

3 Names and Identifiers

3.1 Computed Descriptors

3.1.1 IUPAC Name

(2S)-2-amino-4-(3-carboxypropanoyloxy)butanoic acid
Computed by Lexichem TK 2.7.0 (PubChem release 2021.10.14)
[PubChem](#)

3.1.2 InChI

InChI=1S/C8H13NO6/c9-5(8(13)14)3-4-15-7(12)2-1-6(10)11/h5H,1-4,9H2,(H,10,11)(H,13,14)/t5-m/s1
Computed by InChI 1.0.6 (PubChem release 2021.10.14)
[PubChem](#)

3.1.3 InChIKey

GNISQJGXJIDKDJ-YFKPBYRVSA-N
Computed by InChI 1.0.6 (PubChem release 2021.10.14)
[PubChem](#)

3.1.4 SMILES

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- 12 Interactions and Pathways
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- 15 Information Sources

2. 若是在這一區中沒有 HMDB，就找 Metabolomics Workbench ID

pubchem.ncbi.nlm.nih.gov/compound/439406#section=Metabolomics-Workbench-ID

PubChem O-Succinylhomoserine (Compound)

3.3.6 Metabolomics Workbench ID

50402

▶ Metabolomics Workbench

3.3.7 Nikkaji Number

J1.770.210F

▶ Japan Chemical Substance Dictionary (Nikkaji)

3.3.8 Wikidata

Q27098406

▶ Wikidata

3.4 Synonyms

3.4.1 MeSH Entry Terms

MeSH Entry Terms for O-succinylhomoserine

O-succinylhomoserine

▶ Medical Subject Headings (MeSH)

MeSH Entry Terms for O-succinylhomoserine

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進到 Metabolomics Workbench ID 連結頁面，往下找到 HMDB ID 連結

metabolomicsworkbench.org/data/StructureData.php?RegNo=50402

monkey: O-Succinylhomoserine

ClassyFire superclass: Organic acids and derivatives

ClassyFire class: Carboxylic acids and derivatives

ClassyFire subclass: Amino acids, peptides, and analogues

ClassyFire direct parent: Alpha amino acids

Massbank MS spectra: View MS spectra

NP-MRD NMR spectra: View NMR spectra

SMILES: C(CC(=O)OCC[C@@H](C(=O)O)N)C(=O)O

Studies: Available studies (via RefMet name)

Select appropriate tab below to view additional details:

All Database Links Calculated Properties Human Pathways

External database links:

PubChem CID: 439406

CHEBI ID: 16460

HMDB ID: HMDB0255868

KEGG ID: C01118

BMRB ID: bmse000058

NP-MRD ID(NMR): NP0002606

Plant Metabolite Hub(Pmhub): MS000008317

References

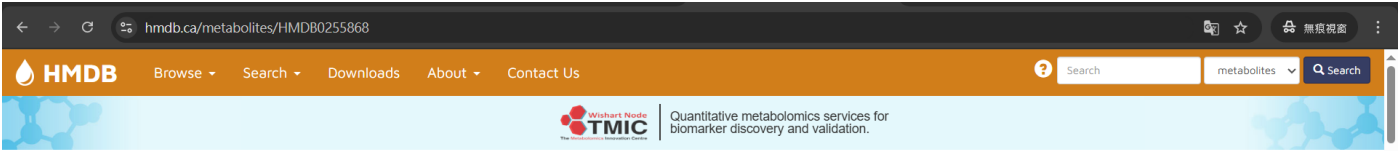
LIPID MAPS classification: "Update of the LIPID MAPS comprehensive classification system for lipids", Fahy E, Subramaniam S, Murphy RC, Nishijima M, Raetz CR, Shimizu T, Spener F, van Meer G, Wakelam MJ, and Dennis EA, J. Lipid Res. (2009) 50: S9-S14. DOI: 10.1194/jlr.R800095-JLR200

ClassyFire classification: "ClassyFire: automated chemical classification with a comprehensive, computable taxonomy", Djoumbou Feunang Y, Eisner R, Knox C, Chepelev L, Hastings J, Owen G, Fahy E, Steinbeck C, Subramanian S, Bolton E, Greiner R, and Wishart DS, J. Cheminformatics (2016) 8:61. DOI: 10.1186/s13321-016-0174-y

NPClassifier classification: "NPClassifier: A Deep Neural Network-Based Structural Classification Tool for Natural Products", Kim HW, Wang M, Leber CA, Nothias LF, Reher R, Kang KB, van der Hooft JJJ, Dorrestein PC, Gerwick WH and Cottrell GW. J Nat Prod. 2021 Nov 26;84(11):2795-2807. DOI: 10.1021/acs.jnatprod.1c00399

若是在這個頁面還是沒找到，就將 tmp2 中的該代謝物的列刪除，進行下一個代謝物名稱的搜尋與整理

複製代謝物名稱



Showing metabocard for O-Succinyl-L-homoserine (HMDB0255868)

Jump To Section: Identification Taxonomy Ontology Physical properties Spectra Biological properties Concentrations Links References XML

enzymes (3) [Show 3 proteins](#) [Show Metabolites with Similar Structures](#)

Record Information	
Version	5.0
Status	Detected but not Quantified
Creation Date	2021-09-11 15:47:00 UTC
Update Date	2022-09-22 17:44:26 UTC
HMDB ID	HMDB0255868
Secondary Accession Numbers	None

Metabolite Identification	
Common Name	O-Succinyl-L-homoserine
Description	Homoserine, O-succinyl- belongs to the class of organic compounds known as alpha amino acids. These are amino acids in which the amino group is attached to the carbon atom immediately adjacent to the carboxylate group (alpha carbon). Based on a literature review very few articles have been published on Homoserine, O-succinyl-. This compound has been identified in human blood as reported by (PMID: 31557052 View). O-succinyl-L-homoserine is not a naturally occurring metabolite and is only found in those individuals exposed to this compound or its derivatives. Technically O-Succinyl-L-homoserine is part of the human exposome. The exposome can be defined as the collection of all the exposures of an individual in a lifetime and how those exposures relate to health. An individual's exposure begins before birth and includes insults from environmental and occupational sources.

到可查詢代謝物 ID 的檔案查詢 ID，並複製該代謝物的 ID

GBM_metabolite_expression_matrix.xlsx - Excel

剪貼簿 剪下 複製 貼上 複製格式

B I U 字體 12 對齊方式 自動換行 跨欄置中

通用格式 設定格式 格式化為表格 表格

一般 中等 好 壞 儲存格 插入 刪除 格式 儲存格 自動加總 填滿 排序與篩選 尋找與擷取

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
74	DMTDB000065	1,2-Cyclohexanedione													
75	DMTDB000066	15-Methylpalmitate													
76	DMTDB000067	(isoheptadecanoic acid)													
77	DMTDB000068	5-Methoxytryptamine													
78	DMTDB000069	Putrescine													
79	DMTDB000070	Tetracosanoic acid													
80	DMTDB000071	(lignoceric acid)													
81	DMTDB000072	Phosphoserine													
82	DMTDB000073	Dodecanoic acid													
83	DMTDB000074	(lauric acid)													
84	DMTDB000075	3-Cyanoalanine													
85	DMTDB000076	(cyano-L-alanine)													
86	DMTDB000077	O-Succinyl-L-homoserine													
87	DMTDB000078	Oleamide													
88	DMTDB000079	sophorose													
89	DMTDB000080	Caffeic acid													
90	DMTDB000081	(3,4-dihydroxycinnamic acid)													
91	DMTDB000082	Glycerophosphoric acid													
92	DMTDB000083	(Glycerol 1-phosphate)													
93	DMTDB000084	(glycerol-alpha-phosphate)													
94	DMTDB000085	Capric acid													
95	DMTDB000086	(Decanoic Acid)													
96	DMTDB000087	Norepinephrine													
97	DMTDB000088	(noradrenaline)													
98	DMTDB000089	atropine***													

尋找及取代 尋找(D) 取代(D) 尋找目標(N): O-Succinyl-L-homoserine 選項(D) >> 全部尋找(D) 找下一個(D) 關閉

到 **modified_result_median** 將 ID 填入第一欄、填入代謝物名稱
(若是從 HMDB 找到的代謝物名稱與 **tmp2** 的代謝物名稱不同，填入 HMDB 的代謝物名稱)

CPTAC3_PDC000546_GBM_metabolite_expression_matrix_byCaseID.xlsx - Excel

檔案常用插入版面配置公式資料校閱檢視開發人員ACROBAT告訴我您想要執行的動作...登入共用

剪下複製貼上複製格式剪貼等

新標題12A⁺A⁻BBIU⁻A⁻中⁻對齊方式

到 **modified_result_max** 中將 ID、代謝物名稱覆蓋過去

CPTAC3_PDC000546_GBM_metabolite_expression_matrix_byCaseID.xlsx - Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
189	DMTDB000029	Pelargonic acid	896900.4	46027.75	471966.4	366013.5	1044402	51530.32	126024.5		86742.43	890125.3	555182.3	165335.2	709666.7	
190	DMTDB000078	Norepinephrine	26745110		29869.44	9536297	8819288	2453467	165494.5		10287091			4666003		
191	DMTDB000031	Nornicotine	563487.8						1977518						923149.8	
192	DMTDB000106	O-Acetylserine														
193	DMTDB000264	Batyl alcohol	5409679													
194	DMTDB000326	Octanal	294125.2		97181.23	253688.4		313705.8			355293.5	192019		52697.96		
195	DMTDB000074	Oleamide			1334580											
196	DMTDB000070	Phosphoserine	1.46E+08													
197	DMTDB000058	Ornithine	18890034			3640219	903227.1		1784919			3490620	1721546	1472500	381190.6	
198	DMTDB000095	O-Acetylserine	5647400		136737.5	25490.66		213481		34231.13	984322.1	2024667	39031.37	49776.19		1995547
199	DMTDB000073	O-Succinyl-L-homoserine	43730.62				13254.59		627397.1			756721.8				
200		oxamic acid														
201		oxoproline														
202		oxoproline						636746.4		358503.3			479130.2			
203		palmitoleic acid	50887.63													
204		parabanic acid NIST	1619345	658888.3	673460.8	172108.7	835875.8	632187.8	8842126	1880458	1573540	203579.5	655932.8	302092.8	296155.1	1407820
205		pelargonic acid	272460.9	633354.5		119488.4		2582849			109125.4					
206		pentadecanoic acid														
207		pentobarbital NIST	132952.5			87146.66	76576.7	78681.48	134815.9	42027.58	55297.32	25067.52	171793.1	50646.02	74038.3	673475.6
208		phenacetic acid 2TMS	7257634	3888.301		12045.09					82890.77				24116.57	
209		phenylacetamide												283339.4	58699.47	119774
210		phenylalanine														
211		phenylalanine minor											168304			
212		phenylethanol NIST														
213		phenylethanol NIST										159857.1				

modified_result_max tmp1 tmp2 modified_result_median_tmp modified_result_median

到 **modified_result_median_tmp** 中輸入公式 =MEDIAN(範圍)

先輸入第一格的公式，範圍是 **tmp2** 中屬於同一個代謝物的列

2					=MEDIAN('tmp2'!B2:B11)				
A					B				
base_name					C243909				
O-Succinyl-L-homoserine					43730.617				

輸入完公式後，選擇第一格的格子，滑鼠移到格子的右下方會出現黑色十字，按住不放並將滑鼠往右方拖到最後一個 Case ID 再將滑鼠放開

BP	BQ	BR	BS	BT	BU	BV	BW	BX	BY	BZ	CA	CB	CC	CD	CE	CF	CG	CH	CI
C846363	C1230738	C1245129	C207624	C229764	C2415597	C289419	C298152	C317832	C383391	C464694	C761370	C827913	C3N-0302	C3N-0344	C3N-0345	C3N-0119	C3N-0119	C3N-01365	
112546.9	#NUM!	405797	255309	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!	289162.9	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!	81975.02	107585.7	#NUM!	#NUM!	

(會出現#NUM 很正常，代表 **tmp2** 欄位範圍內的值都是空值)

複製公式算出的整列中位數，到 **modified_result_median** 中，點選要填入值的第一格，此時點選右鍵，選擇貼上"值"的選項

常用 插入 版面配置 公式 資料 校閱 檢視 開發人員 ACROBAT 告訴我您想要執行什麼

剪下 複製 複製格式 貼上 貼上選項 選擇性貼上(S)... 智慧查閱(L) 插入複製的儲存格(E)... 刪除(D)... 清除內容(N) 快速分析(Q) 篩選(E) 排序(O) 插入註解(M) 儲存格格式(E)... 從下拉式清單挑選(O)... 顯示注音標示欄位(S) 定義名稱(A)... 超連結(H)...

新細明體 12 A A \$ % , B I U 中

A	B	C	D	E	F	G
MTDB000311	Nicotinic acid					
MTDB000037	N-methyl-L-glutamic Acid					
MTDB000008	Nonadecanoic acid	70359				995
MTDB000029	Pelargonic acid	79417				0943 540255.
MTDB000078	Norepinephrine	191398				4.72 275880.
MTDB000031	Nornicotine	56348				6297 881928
MTDB000106	O-Acetylserine					
MTDB000264	Batyl alcohol	27999				
MTDB000326	Octanal	18625				88.4
MTDB000074	Oleamide					
MTDB000070	Phosphoserine	1.46E-				
MTDB000058	Ornithine	136264				2859 903227.
MTDB000025	Orotic acid	34048				0.66
MTDB000073	O-Succinyl-L-homoserine					

新細明體 12 A A \$ % , B I U 中

Ctrl + F -> 取代，將#NUM!取代成空值

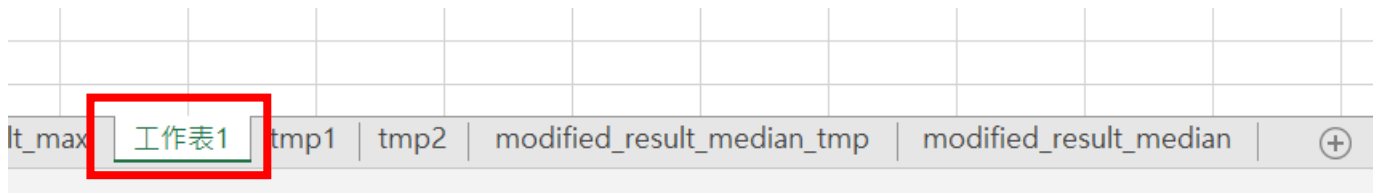
回到 `modified_result_median_tmp` 將剛剛的公式那一列刪除
(每做完一次就刪除保持淨空，避免不同代謝物的值混在一起)

回到 tmp2 將已整理的代謝物刪除

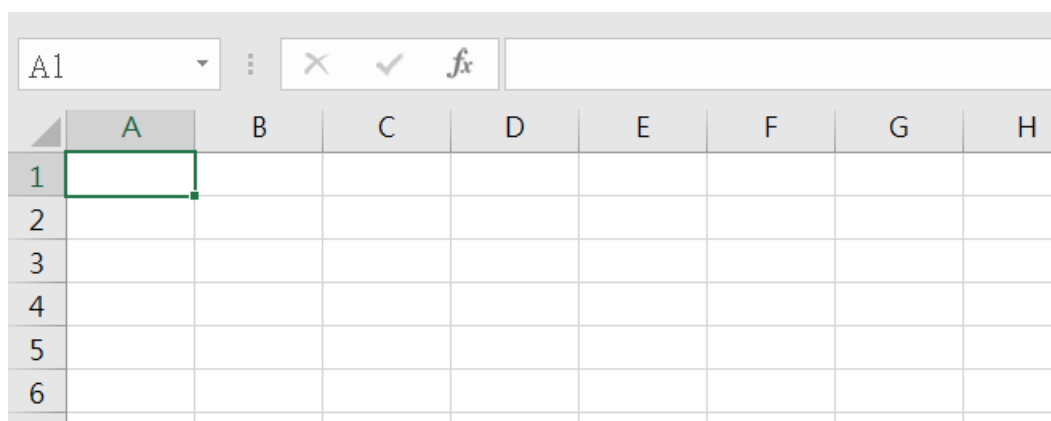
base_name	C243909	C685110	C3N-0348	C3N-03755	C
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine	43730.62				
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine					
O-succinylhomoserine					
o-toluic acid					

若是待整理代謝物的列只有一列，不需要輸入公式的步驟，直接將該列複製到 **modified_result_median**，但是整理代謝物名稱與 ID 的那個步驟不能跳過!!

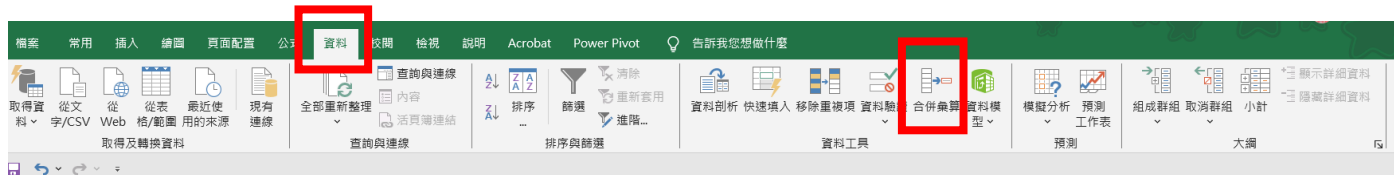
若是在 **modified_result_max** 中的 HMDB 的名稱屬於同一個但有超過 1 個以上的情況，使用 excel 合併彙算功能合併計算最大值
先新開一個暫時的工作表



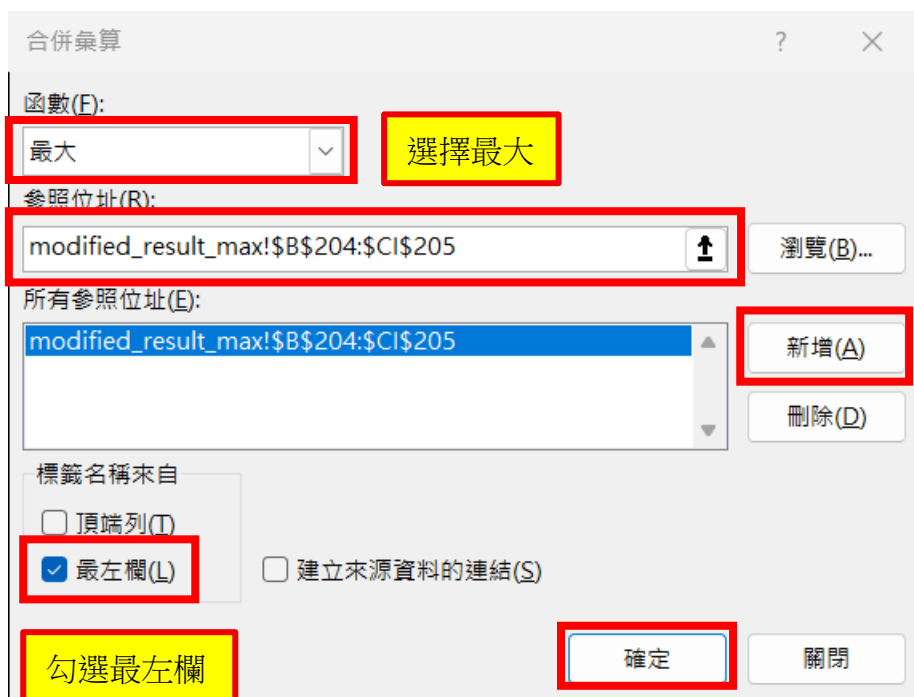
選擇第一格



”資料” -> “合併彙算”



記得在要選取合併計算最大值前，先把屬於同一個代謝物但不同名稱改成同一個代謝物名稱!!



點選右方黑色 icon
切換下方工作表，選取
要取最大值的範圍，
選完後按”新增”

將合併彙算後的結果複製貼到 **modified_result_max** 的同個代謝物名稱的第一列，並刪除下方同名稱的代謝物的列，再刪除新增的工作表(記得每做一次合併彙算就新增一次新的工作表，因為使用舊的工作表有時候跑出來的結果會很奇怪，所以開新的工作表做合併彙算比較保險)

在 **tmp2** 將已整理的代謝物刪除後，即可進行下一個代謝物名稱的整理